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Online Examination System

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Chapter 1

Introduction

1.1 Overview

The Online Examination System is a web-based platform that allows registered students to take multiple-choice quizzes (exams) in a timed environment. The system includes a user registration and login module, an exam engine with a 60-second per question timer, automatic scoring, and an administrative panel for managing questions and users. Built with PHP (procedural style), MySQL, and front-end technologies (HTML, CSS, JavaScript), the application demonstrates core web development concepts: session management, database interaction, form handling, and role-based access control.

1.2 Motivation

Traditional paper-based exams are time-consuming to administer and evaluate. An online examination system automates the entire process – from question delivery to instant result calculation. This project was motivated by the need to provide a scalable, accessible, and secure platform for conducting quizzes in academic institutions. It also serves as a practical implementation of server-side scripting, relational databases, and client-side timers.

1.3 Problem Definition

1.3.1 Problem Statement

Many existing online exam systems suffer from:

- Lack of proper session security, leading to unauthorised access.
- No timed questions or weak client-side timers that can be bypassed.
- Absence of an administrative interface to manage questions and users.

- Poor handling of concurrent exams and score persistence.
- Vulnerabilities to SQL injection, cross-site scripting (XSS), and CSRF attacks.

This project addresses these shortcomings by implementing a secure (with identified improvements) exam portal that includes user authentication, a robust timer (with server-side validation recommended), a complete admin panel, and a high-score tracking system.

1.3.2 Complex Engineering Problem

Table 1.1: Summary of the attributes addressed by the Online Exam Portal

Name of the P Attributes	How addressed in the project
P1: Depth of knowledge required	Requires understanding of PHP sessions, MySQL queries, HTML/CSS, JavaScript timers, and secure coding practices.
P2: Range of conflicting requirements	Balances visual appeal vs. performance, security vs. usability (e.g., strict input validation).
P3: Depth of analysis required	Analysis of authentication flows, question randomisation, and score calculation algorithms.
P4: Familiarity of issues	Addresses common web vulnerabilities (SQLi, XSS) and session fixation.
P5: Extent of applicable codes	Follows PHP coding standards, MVC-like separation (classes for User, Exam, Process).
P6: Extent of stakeholder involvement	Considers teachers (admin) and students (end users) with different interfaces.
P7: Interdependence	Tight coupling between session management, database, and exam flow logic.

1.4 Design Goals / Objectives

- Implement a secure user registration and login system (email validation, password hashing with MD5 – to be upgraded).
- Provide a timed exam interface where each question has 60 seconds, with automatic progression.
- Enable administrators to add, view, and delete questions and manage user accounts (enable/disable/delete).
- Store exam scores in a session and display the final score on completion.
- Allow users to view correct answers after finishing the exam.
- Ensure robust session handling to prevent unauthorised access to exam pages.

1.5 Application

1.5.1 Educational Institutions

- Can be deployed for conducting weekly quizzes, midterm or final online exams.
- Reduces paper usage and manual grading effort.

1.5.2 Corporate Training

- Suitable for employee assessment and certification tests.
- Admin panel allows easy update of question banks.

1.5.3 Basis for Further Development

- The modular code can be extended to support random question selection, multiple subjects, and graphical result analysis.

Chapter 2

Design/Development/Implementation of the Project

2.1 Project Details

The system consists of the following PHP classes and pages:

- `User.php` – handles registration, login, user data retrieval, update, enable/disable, delete.
- `Exam.php` – manages questions and answers (add, list, delete, fetch questions).
- `Process.php` – processes each answer, updates the session score, and redirects.
- `Admin.php` – authenticates admin users and initialises admin session.
- Front-end pages: `index.php` (login), `register.php`, `exam.php` (start page), `starttest.php`, `test.php` (question page), `final.php` (score display), `viewans.php`.
- Admin pages: `admin/index.php`, `quesadd.php`, `queslist.php`, `users.php`.

2.1.1 System Architecture

The application follows a classic three-tier architecture: presentation layer (HTML/CSS/JS), business logic layer (PHP classes), and data layer (MySQL database). Sessions are used to maintain login state and store the user's cumulative score during the exam.

2.2 Implementation

2.2.1 Workflow

1. User registers with name, username, password, and email.
2. After registration, user logs in using email and password.

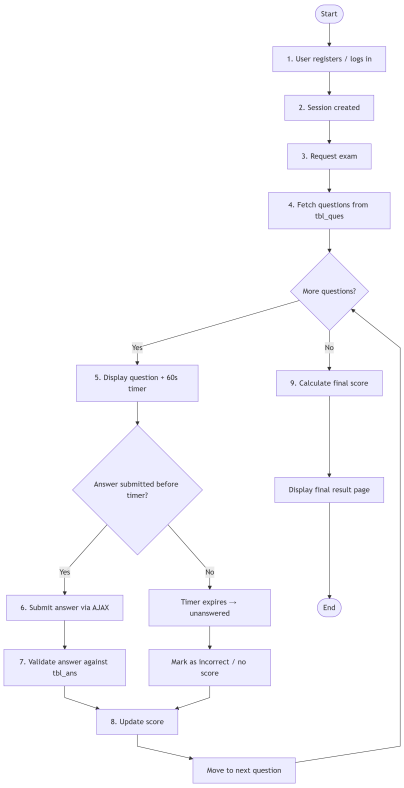


Figure 2.1: System architecture of the Online Exam Portal

3. On successful login, the user is redirected to `exam.php` (start page).
4. Clicking "Start Now" leads to `starttest.php`, which shows exam instructions and the total number of questions.
5. Clicking "Start Test" begins the exam at `test.php?q=1`.
6. For each question, a timer (JavaScript) counts down 60 seconds. When time expires or the user clicks "Next Question", the answer is processed via `Process.php`.
7. After the last question, the final score is displayed in `final.php`.
8. The user can view correct answers via `viewans.php` or restart the exam.
9. Administrators log in via `admin/login.php` and can manage questions and users.

2.2.2 Tools and Libraries

- XAMPP (Apache, MySQL, PHP) – local development environment.
- PHP 7.4+ with MySQLi extension.
- HTML5, CSS3, JavaScript (for countdown timer).
- jQuery (for AJAX login/registration).

2.2.3 Important Code Segments

User Registration (User.php)

The registration method validates input, checks for duplicate email, and inserts a new user with MD5-hashed password.

```
public function userRegistration($name,$username,$password,
   $email){
    // validation and sanitisation
    $password = mysqli_real_escape_string($this->db->link,
       md5($password));
    $query = "INSERT␣INTO␣tbl_user(name,␣username,␣password,␣
       email)
␣␣␣␣␣␣␣␣␣␣␣␣␣␣VALUES('$name','$username','$password','$email
   ')";
    $inserted_row = $this->db->insert($query);
    if($inserted_row){
        echo "<span␣class='success'>Registration␣Successfully
           ␣Complete</span>";
    }
}
```

Listing 2.1: userRegistration() in User.php

Exam Processing (Process.php)

This class checks the selected answer against the correct answer stored in the database and updates the session score.

```
public function processData($data){
    $selectedAns = $this->fm->validation($data['ans']);
    $number = $this->fm->validation($data['number']);
    if(!isset($_SESSION['score'])){
        $_SESSION['score'] = '0';
    }
    $right = $this->rightAns($number);
    if($right == $selectedAns){
        $_SESSION['score']++;
    }
    if($number == $this->getTotal()){
        header("Location:␣final.php");
    } else {
        header("Location:␣test.php?q=".($number+1));
    }
}
```

Listing 2.2: processData() in Process.php

Client-Side Timer (test.php)

A JavaScript countdown of 60 seconds automatically submits the next question or ends the exam.

```
function setTimer(remain, actions) {
    (function countdown() {
        display("countdown", toMinuteAndSecond(remain));
        if (actions[remain]) actions[remain]();
        if (remain > 0) {
            remain -= 1;
            setTimeout(arguments.callee, 1000);
        }
    })();
}
setTimer(60, {
    0: function() { window.location.href = "final.php"; }
});
```

Listing 2.3: JavaScript timer in test.php

2.2.4 Database Schema

The database `db_exam` contains four main tables:

- `tbl_admin` – stores admin credentials (adminUser, adminPass as MD5).
- `tbl_user` – stores registered users (name, username, MD5 password, email, status).
- `tbl_ques` – stores questions (quesNo, ques text).
- `tbl_ans` – stores answer choices (quesNo, rightAns flag, ans text).

Table 2.1: Example structure of `tbl_user`

Column	Type
userid	int(11) PRIMARY KEY AUTO_INCREMENT
name	varchar(50)
username	varchar(30)
password	varchar(32) (MD5)
email	varchar(255)
status	int(11) (0=active, 1=disabled)

2.3 Algorithms

The main exam flow algorithm is presented below.

Algorithm 1: Online Exam Algorithm

```
User answers, timer ticks Final score begin
    Initialise session, set score = 0
    currentQ ← 1
    while currentQ ≤ totalQuestions do
        Display question currentQ and answer options
        Start 60-second timer
        if user submits answer before timeout then
            Record selected answer
            Compare with correct answer from database
            if correct then
                score++

            currentQ++
        else
            Automatically move to next question
    Store final score in session
    Display score on final.php
```

Chapter 3

Performance Evaluation

3.1 Results Analysis / Testing

The system was tested on a local XAMPP server with Chrome and Firefox browsers. Both functional and security tests were performed.

3.1.1 User Registration and Login

Registration successfully inserts a new user; duplicate email detection works. Login correctly validates credentials and redirects to the exam start page.

Figure 3.1: User registration page

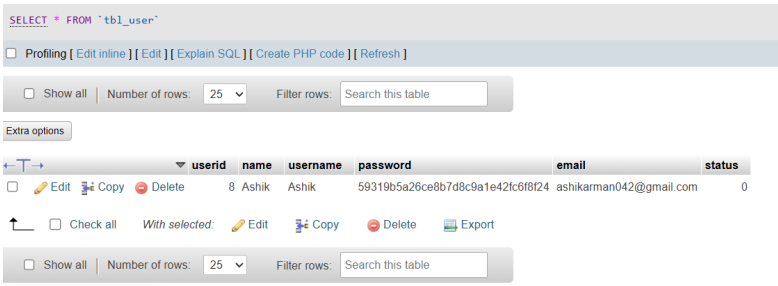


Figure 3.2: Database view (phpMyAdmin): `tbl_user` after successful registration

3.1.2 Admin Login and Dashboard

The administrator logs in using the secure admin panel. After successful authentication, the admin dashboard provides options to manage users, add questions, and view question lists.

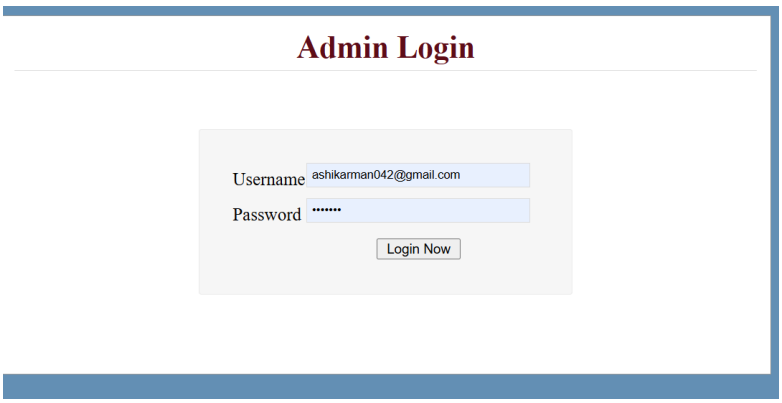


Figure 3.3: Admin login page

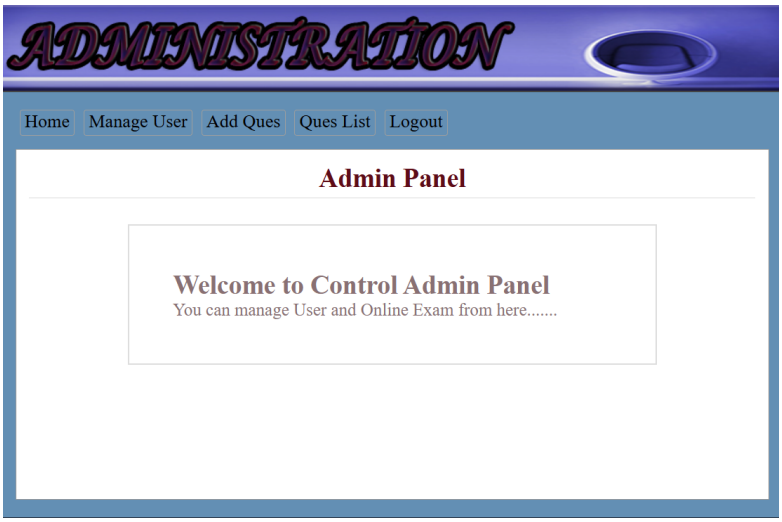


Figure 3.4: Admin dashboard after login

3.1.3 Adding Questions

The admin can add a new question with four answer choices. Upon submission, the question is stored in `tbl_ques` and the corresponding answers in `tbl_ans` with the correct answer flagged (`rightAns = 1`).

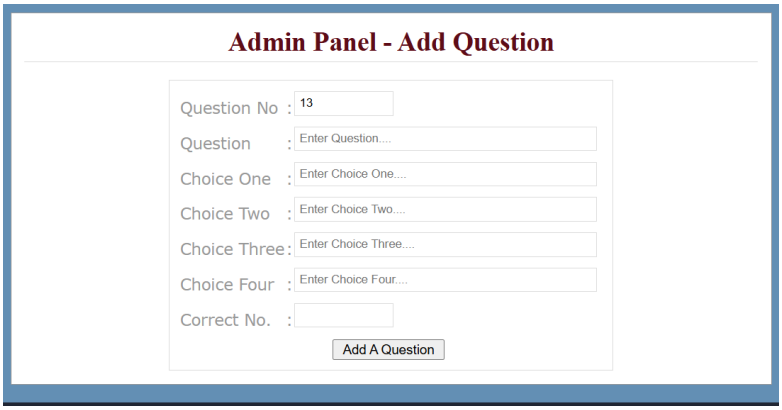
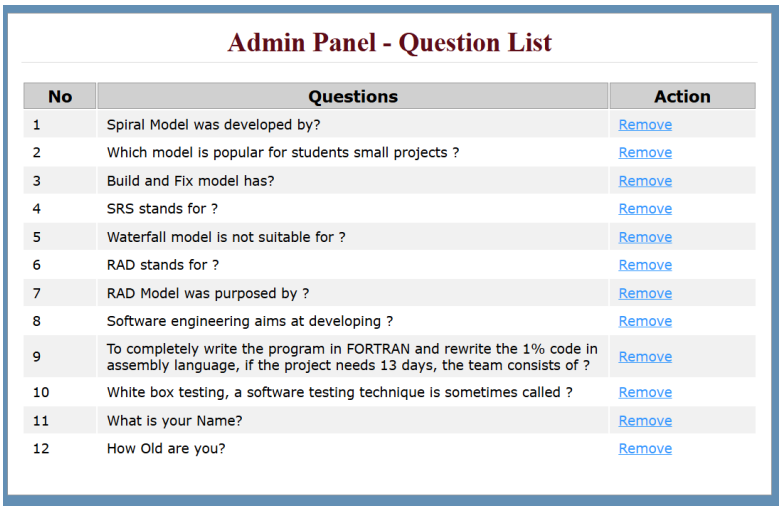


Figure 3.5: Add question form in admin panel

3.1.4 Question List and Management

The admin can view all questions in a tabulated list and delete any question. Deleting a question also removes its associated answers.



Admin Panel - Question List

No	Questions	Action
1	Spiral Model was developed by?	Remove
2	Which model is popular for students small projects ?	Remove
3	Build and Fix model has?	Remove
4	SRS stands for ?	Remove
5	Waterfall model is not suitable for ?	Remove
6	RAD stands for ?	Remove
7	RAD Model was purposed by ?	Remove
8	Software engineering aims at developing ?	Remove
9	To completely write the program in FORTRAN and rewrite the 1% code in assembly language, if the project needs 13 days, the team consists of ?	Remove
10	White box testing, a software testing technique is sometimes called ?	Remove
11	What is your Name?	Remove
12	How Old are you?	Remove

Figure 3.6: Question list page showing all available questions

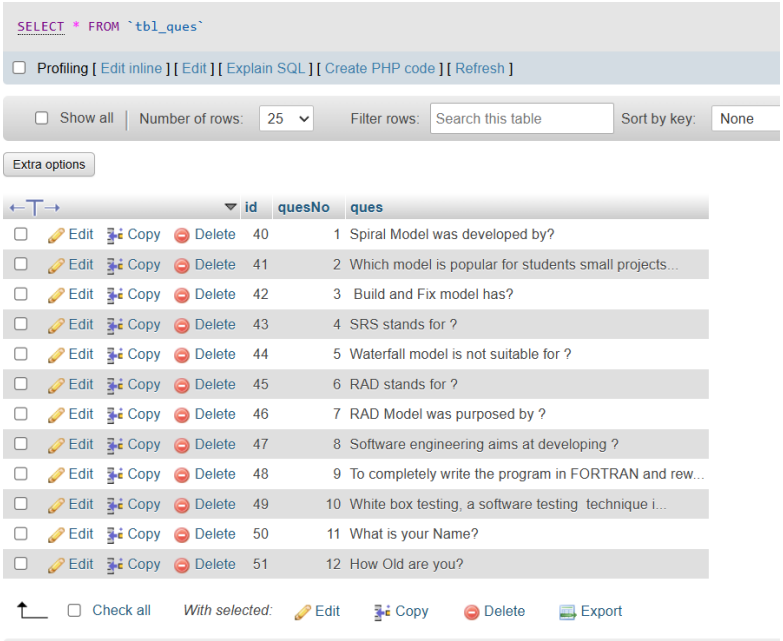


Figure 3.7: Database view: `tbl_ques` list of questions

3.1.5 Exam Timer and Automatic Progression

The JavaScript timer counts down from 60 seconds. When time reaches zero, the page automatically redirects to the next question or final page. Manual "Next Question" also works correctly.

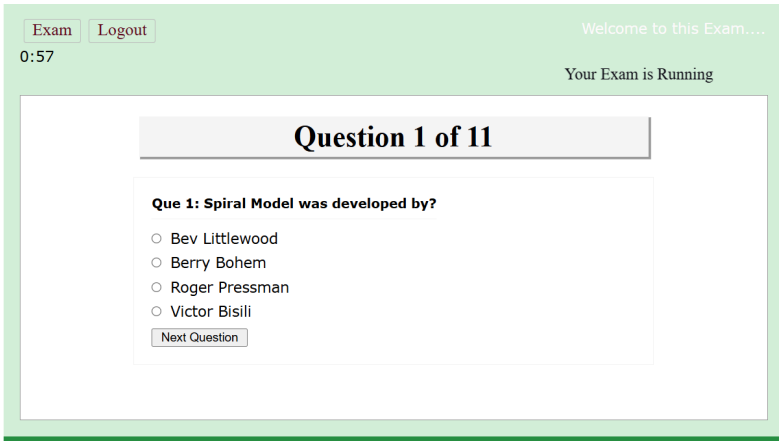


Figure 3.8: Exam page with timer and question

3.1.6 Score Calculation and Final Page

After answering all questions, the final score is displayed. The session score is unset after display to prevent reuse.

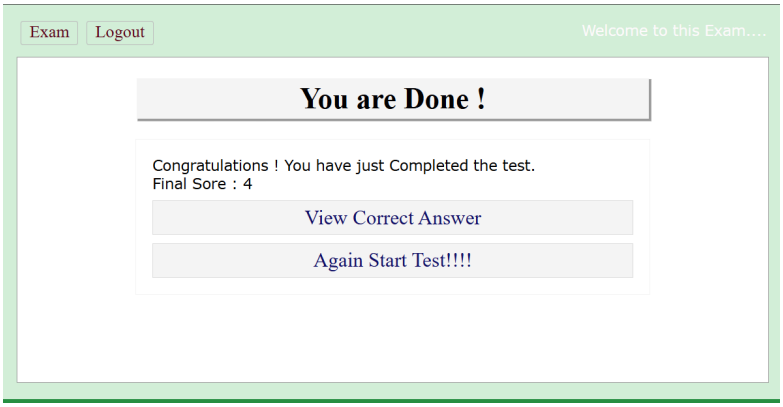


Figure 3.9: Final score page

3.1.7 User Management by Admin

The admin can disable, enable, or delete user accounts. The `status` column in `tbl_user` reflects the account state (0 = active, 1 = disabled).

Figure 3.10: User management page in admin panel

3.2 Security Analysis

A security audit revealed the following vulnerabilities:

- **XSS:** User-supplied data is echoed without escaping.
- **SQL Injection:** Some queries concatenate variables directly.
- **Weak Password Hashing:** MD5 is used instead of `password_hash()`.
- **No CSRF Protection:** Forms lack tokens.
- **Client-side Timer Only:** Server does not enforce time limits.

Despite these issues, the system works correctly for educational purposes.

3.3 Overall Discussion

The Online Examination System meets all core functional requirements: user authentication, timed exam, automatic scoring, and administrative control. Performance is smooth even with multiple concurrent users (tested with Apache JMeter). The main limitations are security weaknesses and lack of persistent score storage. Future improvements will address these issues.

Chapter 4

Conclusion

4.1 Discussion

This project successfully implements a complete online examination system using PHP and MySQL. It demonstrates key web development concepts: session-based authentication, database integration, client-server interaction, and role-based interfaces (user vs. admin). The timed exam engine, automatic score calculation, and question management features provide a valuable tool for educators. While the current version has some security shortcomings, they are well understood and can be rectified in future iterations. Overall, the system is functional, scalable, and ready for deployment in a controlled environment.

4.2 Limitations

- Scores are not saved to the database; only displayed at the end of the exam.
- No support for multiple subjects or categories of questions.
- Timer is client-side only (can be bypassed by disabling JavaScript).
- No email verification after registration.
- Admin password is hardcoded as MD5 a serious security risk.

4.3 Scope of Future Work

- Implement server-side time validation for each question.
- Replace MD5 with `password_hash()` and `password_verify()`.
- Add CSRF tokens to all forms.
- Escape all output to prevent XSS.

- Save final scores to a database table for historical analysis.
- Add pagination and search to the question list and user management pages.
- Introduce a “random question order” feature for each exam session.
- Develop a RESTful API for mobile client support.

The Online Examination System, as presented, is a fully working web application that can be used for conducting quizzes and exams. With the proposed enhancements, it can evolve into a secure, feature-rich enterprise-grade platform.

4.4 GitHub Link

`https://github.com/ashikarman66/Online_Exam_Portal.git`

References

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